

MASTER OF BUSINESS ADMINISTRATION (MBA)

Examination Paper Solutions: Management Information Systems & E-Governance (CC-105)

Course Code: CC-105 /
MB-1011

Subject: Management Information Systems & E-
Governance

Academic Year: 2022

Time Allowed: 3 Hours

Full Marks: 70

Status: Master Answer Key

GROUP — A (COMPULSORY QUESTIONS)

Question 1: Answer all questions within one or two sentences.

5 × 1 = 5 Marks

(a) Application Software

Application software refers to a subclass of computer programs designed to perform specific user-oriented tasks, such as word processing, spreadsheet calculations, or database management, running on top of the system software.

(b) Secondary Storage

Secondary storage consists of non-volatile, long-term memory devices, such as hard drives, solid-state drives, and optical discs, which retain digital data permanently even when the computer system is powered down.

(c) Information

Information is raw data that has been systematically processed, organized, structured, and contextualized to deliver meaningful insights and support effective decision-making within an enterprise.

(d) Big Data

Big Data represents massive, complex, and rapidly expanding datasets characterized by high volume, high velocity, and high variety that surpass the processing capabilities of traditional relational database management systems.

(e) Read Only Memory (ROM)

Read Only Memory (ROM) is a type of built-in, non-volatile semiconductor memory that permanently stores essential system startup instructions, such as the BIOS, which can be read but not modified during standard operations.

Question 2: Answer all questions within four or five sentences.

2.5 × 2 = 5 Marks

(a) What is closed source?

Closed source software, or proprietary software, refers to digital programs whose underlying source code remains hidden from the general public and is legally restricted under copyright law. The intellectual property is strictly owned by the creator or vendor, who grants users access through restrictive licensing agreements without the right to modify or redistribute it. Users are entirely dependent on the vendor for software updates, technical security patches, and structural feature upgrades. Common enterprise examples include Microsoft Windows, Oracle Databases, and Adobe Creative Cloud suites.

(b) What are Transaction Processing Systems (TPS)?

A Transaction Processing System (TPS) is a foundational enterprise information system that captures, processes, records, and stores the structured data generated during day-to-day business operations. Operating at the operational level of management, a TPS handles high-volume, repetitive activities such as sales orders, payroll processing, bank deposits, and airline flight bookings. It guarantees data integrity by enforcing ACID properties (Atomicity, Consistency, Isolation, Durability) across all concurrent system entries. By automating routine workflows, a TPS feeds critical real-time data to higher-level management systems like MIS and DSS.

GROUP — B (LONG ANSWER QUESTIONS)

Note: According to examination instructions, answer any four questions from this group. For complete comprehensive preparation, all questions (Q3 through Q8) have been solved below matching the requested academic word counts.

Question 3:**15 Marks**

- (a) Discuss how MIS functions as a Corporate Resource.
- (b) Write a short note on Electronic Data Processing (EDP).

(a) Discuss how MIS functions as a Corporate Resource

In the contemporary business paradigm, data and information have evolved from administrative byproducts into strategic assets equivalent to traditional physical resources like capital, land, and human labor. A Management Information System (MIS) serves as the primary structural mechanism that transforms this raw data into a structured corporate resource. By systematically integrating hardware, software, telecommunications, and human workflows, an MIS builds a comprehensive digital foundation that captures internal and external operational dynamics. It democratizes critical information across business hierarchies, breaking down departmental silos and ensuring that organizational leaders can execute evidence-based decisions swiftly. Consequently, MIS functions as an invaluable corporate asset by optimizing resource deployment, enhancing internal operational efficiencies, and creating sustained competitive advantages.

The modern operational environment demands continuous, cross-functional visibility to remain agile. MIS satisfies this requirement by providing distinct vertical and horizontal data streams throughout the enterprise. For operational managers, it generates structured exception reports, tracking daily manufacturing outputs or real-time sales inventories against predefined benchmarks. For middle management, MIS offers comparative analytical summaries that help evaluate quarterly regional market performances or allocate corporate budgets. At the executive tier, it aggregates highly summarized macro-level dashboards that align business unit performance with long-term corporate governance objectives. By establishing a single, undisputed version of organizational truth, MIS drastically reduces transaction costs, minimizes processing friction, and empowers management to respond proactively to shifting regulatory frameworks and market conditions.

Furthermore, the value of MIS as a resource is underscored by its ability to drive strategic corporate differentiation. Through automated forecasting modules, supply chain optimization algorithms, and customer relationship analytics, MIS reveals hidden revenue opportunities and operational inefficiencies. It allows enterprises to transition from a reactive posture to a predictive operational model. This capability lowers overhead, enhances customer satisfaction metrics, and shortens product times-to-market. Ultimately, treating MIS as a core corporate resource transforms IT investments from simple cost centers into strategic engines of organizational value, scalability, and long-term business resilience.

(b) Write a short note on Electronic Data Processing (EDP)

Electronic Data Processing (EDP) represents the foundational computerized processing of transactional data using digital computer networks, storage units, and specialized business applications. Originating as a tactical automation mechanism for high-volume clerical activities, EDP focuses on handling basic, structured corporate transactions with maximum speed and minimal human error. It involves a systematic data cycle: capturing raw source documents, validating records, sorting data, performing mathematical or logical calculations, and generating structured digital outputs. Unlike comprehensive decision-oriented systems like MIS, EDP primarily targets operational efficiency, transactional accuracy, and the reduction of manual administrative overhead.

The fundamental core of an EDP infrastructure relies on its ability to process data through either batch processing or real-time online processing methods. In batch processing, transactions are aggregated over a specific timeframe and run as a single job block, which is ideal for predictable activities like monthly corporate payroll or utility billing cycles. In contrast, online real-time processing handles data updates instantly as they occur, supporting dynamic business operations such as banking ATM withdrawals and retail point-of-sale systems. By standardizing file management and replacing manual ledgers with structured digital databases, EDP establishes the required data integrity, scalability, and technical baseline that enables more sophisticated enterprise information architectures to exist.

Question 4:**15 Marks**

- (a) What are Programming Languages? Differentiate between Compilers and Interpreters with suitable examples.
- (b) Write a short note on the different generations of computers.

(a) What are Programming Languages? Differentiate between Compilers and Interpreters

A programming language is a highly structured, formal vocabulary and set of grammatical rules used to construct instructions that a computer system can interpret and execute. It acts as the primary translator between human logic and the binary operational realities of computer hardware. These languages range from low-level variants closely tied to central processing unit architectures, such as machine code and assembly, to sophisticated high-level options like Java, Python, and C++. High-level languages use abstract, English-like syntax to simplify code development, make software maintainable, and enable cross-platform portability. However, because CPUs can only process binary code, computer systems require language processors—specifically compilers and interpreters—to convert high-level instructions into machine-readable formats.

Parameter	Compiler	Interpreter
Execution Mechanism	Translates the entire high-level source code file into machine code at once, generating a standalone executable file before run time.	Translates and executes the source code line-by-line, on the fly, during runtime without creating an intermediate binary file.
Processing Speed	Initial compilation is slower, but the resulting executable runs much faster since no further translation occurs.	Starts executing immediately without compilation delays, but overall runtime is slower due to ongoing translation.
Error Diagnosis	Scans the full program and reports all syntax and structural errors at the end of compilation, making early debugging challenging.	Stops execution the precise moment it encounters an error, making it highly interactive and easier to debug code.
Memory Utilization	Requires more memory to store the newly generated intermediate object code and executable files.	More memory efficient because it translates code incrementally without saving separate executable files.
Industry Examples	Used by performance-critical languages like C , C++ , and Rust .	Used by dynamic, flexible languages like Python , Ruby , and JavaScript .

(b) Write a short note on the different generations of computers

The evolutionary timeline of electronic computing is categorized into five distinct technological generations, each defined by a major breakthrough in core electronic components. The **First Generation (1940–1956)** relied on vacuum tubes for circuitry and magnetic drums for internal memory. These systems, like ENIAC and UNIVAC, were massive, consumed immense amounts of electricity, generated heavy heat, and required primitive machine language instructions. The **Second Generation (1956–1963)** marked a major shift by replacing fragile vacuum

tubes with transistors. This upgrade dramatically reduced computer size, increased operational speeds, and enabled the creation of early assembly and high-level languages like FORTRAN and COBOL.

The **Third Generation (1964–1971)** advanced the industry further through the invention of Integrated Circuits (ICs), which packed hundreds of transistors onto a single silicon semiconductor chip. This era introduced keyboards, monitors, early operating systems, and basic multi-programming environments. The **Fourth Generation (1971–Present)** emerged with the development of Large Scale (LSI) and Very Large Scale Integration (VLSI), compressing millions of components onto a single microchip processor. This technology gave rise to personal computing (PCs), portable laptops, localized networking, and the modern internet. Finally, the **Fifth Generation (Present & Beyond)** centers on Artificial Intelligence (AI), Ultra Large Scale Integration (ULSI), quantum computing, and parallel processing frameworks. This generation aims to build machines capable of natural language processing, self-learning, and complex autonomous reasoning.

Question 5:

15 Marks

- (a) Write brief notes on the concept, use and applications of Simon's framework.
- (b) How is Procedural Programming different from Object Oriented Programming? Elucidate with suitable examples.

(a) Write brief notes on the concept, use and applications of Simon's framework

Developed by Nobel Laureate Herbert A. Simon, Simon's framework is a foundational three-stage conceptual model used in management information systems to analyze and map human decision-making processes. The model breaks decision-making down into three sequential phases: **Intelligence, Design, and Choice**. The **Intelligence Phase** focuses on identifying problems or opportunities within the business environment. This involves continuous data collection, scanning operational metrics, and flagging deviations from corporate targets. The **Design Phase** centers on formulating, developing, and analyzing alternative courses of action. Here, managers use modeling techniques and simulation tools to evaluate the feasibility of various solutions. Finally, the **Choice Phase** involves critically evaluating the proposed alternatives and selecting the optimal path for implementation, backed by corporate resources. Later iterations added an **Implementation and Review** phase to monitor outcomes.

In corporate practice, Simon's framework provides a structured blueprint for designing Decision Support Systems (DSS) and Executive Information Systems (EIS). For instance, in enterprise inventory management, the Intelligence phase uses automated systems to detect when raw materials drop below critical thresholds. The Design phase then generates automated purchasing options, balancing vendor pricing, shipping times, and holding costs. During the Choice phase, procurement managers evaluate these choices and sign off on the ideal vendor contract. By breaking complex business issues down into structured steps, Simon's framework helps organizations align their software tools with human cognitive workflows, turning chaotic business data into systematic, auditable corporate choices.

(b) How is Procedural Programming different from Object Oriented Programming?

Procedural Programming and Object-Oriented Programming (OOP) are two contrasting coding paradigms that shape how software systems are built and organized. Procedural programming follows a top-down approach centered on step-by-step instructions, routines, and functions. Code and data are treated as separate entities, and data flows freely across global functions, which can make it hard to maintain as systems scale. In contrast, OOP uses a bottom-up structure that organizes code around 'objects'—self-contained units that bundle both data (attributes) and behavior (methods) together. OOP relies on four core principles: encapsulation (hiding internal data states), inheritance (reusing code across classes), polymorphism (allowing a single interface to represent different actions), and abstraction (hiding complex internal logic).

To illustrate the difference, consider a corporate banking application. In a procedural language like C, managing an account requires passing external variables like `account_number` and `balance` into independent, standalone functions such as `withdraw()` or `deposit()`. This exposes the core financial data to accidental modification elsewhere in the codebase. In an object-oriented language like Java, the system models a clean `BankAccount` class. This class encapsulates private data fields for the account balance and exposes secure, built-

in methods like `account.deposit(amount)`. This structure protects data integrity, isolates bugs, and allows developers to scale complex enterprise systems cleanly.

Question 6:**15 Marks**

- (a) How is good governance achieved through E-Governance?
- (b) Discuss few of the prominent national and global trends witnessed in the implementation of E-Governance policies.

(a) How is good governance achieved through E-Governance

Good governance centers on transparency, accountability, responsiveness, equity, and the rule of law within public administration. E-Governance is a powerful tool for achieving these goals, utilizing information and communication technologies (ICT) to fundamentally transform how public services are delivered, managed, and audited. By shifting from slow, paper-based bureaucratic workflows to automated digital platforms, E-Governance eliminates the subjective delays and gatekeeping that historically fueled public corruption. It builds direct, frictionless communication channels between government bodies, citizens (G2C), and businesses (G2B), ensuring public services are accessible regardless of geographic or socio-economic barriers.

A key mechanism for driving transparency through E-Governance is the real-time publishing of state data, public budgets, and procurement tenders on accessible web portals. This open model allows citizens, media outlets, and civil society organizations to track public spending directly, which significantly increases civic accountability. Additionally, automated workflows—such as online passport processing, digital land registry updates, and electronic tax filing systems—remove intermediaries from service delivery, reducing bribery risks and speeding up processing times. E-Governance also secures data integrity through centralized databases, ensuring public benefits reach intended recipients directly without leakages. Ultimately, by digitizing administrative workflows, governments transform their operations from rigid, opaque bureaucracies into citizen-centric, data-driven institutions that strengthen public trust and democratic engagement.

(b) Discuss prominent national and global trends witnessed in E-Governance

The global landscape of public administration is experiencing a major shift, driven by advanced digital technologies that reshape how states operate. Nationally, India's **Digital India** initiative stands out as a powerful example of large-scale digital transformation. Central to this strategy is the **JAM Trinity** (Jan Dhan accounts, Aadhaar biometric identification, and Mobile connectivity), which has revolutionized public welfare through Direct Benefit Transfers (DBT). This integrated infrastructure allows the state to send financial aid straight to vulnerable citizens, wiping out leakages and systemic corruption. Further advancements, like the Unified Payments Interface (UPI) for financial inclusion, DigiLocker for paperless document management, and national single-window portals for business licensing, show a clear trend toward unified, high-volume digital ecosystems.

Globally, leading nations are pushing the boundaries of citizen-centric public service models. Estonia has established itself as a digital leader through its **e-Estonia** initiative, running 99% of its public services online via a secure decentralized data exchange layer called X-Road. The country pioneered e-Residency and secure blockchain-backed internet voting, setting a global standard for digital statehood. At the same time, international public policies are shifting toward cloud-first strategies, open-data portals, and the integration of Artificial Intelligence to automate complex regulatory compliance. There is also a strong global focus on strengthening national cybersecurity frameworks to safeguard critical infrastructure against cross-border threats, ensuring digital governance models remain secure, resilient, and inclusive as they evolve.

Question 7:

15 Marks

- (a) What are functional information system? Enumerate the key features of Personnel and HR Information Systems.
- (b) Write a short note on Software Development Life Cycle (SDLC).

(a) What are functional information systems? Enumerate key features of Personnel and HR Information Systems

Functional Information Systems are specialized enterprise information architectures designed to support a specific business function or department within an organization. Unlike cross-functional platforms like Enterprise Resource Planning (ERP) systems, these systems tailor their data models, tracking tools, and reporting workflows to the unique operational and strategic needs of distinct business units, such as Finance, Marketing, Production, or Human Resources. By capturing and managing deep departmental data, functional systems streamline day-to-day operations and provide managers with specialized reports to track performance against specific targets.

A Personnel and Human Resource Information System (HRIS) is a vital functional information system built to manage an organization's human capital from recruitment to retirement. Key capabilities of a modern HRIS include:

- **Automated Recruitment & ATS:** Manages job postings, tracks applicant resumes, schedules interviews, and simplifies candidate onboarding workflows.
- **Time & Attendance Tracking:** Monitors employee shift schedules, clock-in times, absences, and leave requests, integrating directly with payroll.
- **Payroll Processing:** Computes salaries, tax withholdings, insurance deductions, and regulatory compliance requirements automatically.
- **Performance & Talent Management:** Records employee performance evaluations, sets professional goals, and tracks training certifications and skill development.
- **Strategic Workforce Analytics:** Generates demographic reports, turnover insights, and workforce planning forecasts to guide executive talent strategies.

(b) Write a short note on Software Development Life Cycle (SDLC)

The Software Development Life Cycle (SDLC) is a structured, phased framework used by IT organizations to engineer, build, deploy, and maintain high-quality software systems cost-effectively. Following a systematic engineering approach, the SDLC reduces project risks, manages development costs, and ensures final software deliverables align with complex business requirements. The standard cycle moves through several distinct phases: **Requirement Analysis** (gathering and documenting stakeholder business needs), **System Design** (creating architectural plans and database models), **Implementation / Coding** (writing clean, standardized code), **Testing** (finding and fixing bugs against quality metrics), **Deployment** (releasing the software to production environments), and **Maintenance** (applying continuous updates and security patches).

Organizations choose from various SDLC methodologies based on their specific project scope, timelines, and stability requirements. The classic **Waterfall Model** follows a strict, linear path where each phase must finish

completely before the next begins, making it a strong choice for well-defined, highly regulated projects. Conversely, modern **Agile Methodologies** focus on iterative, incremental development, breaking projects down into flexible 30-day sprints to adapt quickly to changing user requirements. By enforcing clear documentation benchmarks, rigorous testing standards, and milestone reviews at every stage, the SDLC framework serves as a critical management tool for delivering dependable software solutions that meet or exceed enterprise expectations.

Question 8:

15 Marks

- (a) What are Decision Support Systems (DSS)? How are they different from Expert Systems or Executive Information Systems (EIS)?
- (b) Write brief notes on Data Mining, Augmented Reality and Enterprise Resource Planning (ERP).

(a) What are Decision Support Systems (DSS)? Differentiate from Expert Systems and EIS

A Decision Support System (DSS) is an interactive, computer-based information platform that combines data analytical models, operational databases, and user-friendly interfaces to help managers navigate semi-structured or unstructured decision-making problems. Operating primarily at the tactical management level, a DSS does not replace human judgment; instead, it empowers decision-makers to test scenarios through advanced tools like what-if calculations, sensitivity analysis, and multi-variable optimization modeling. By pulling data from both internal systems and external market feeds, a DSS gives managers the ability to model complex business problems, explore potential outcomes, and choose the most effective strategy.

While a DSS helps managers explore options, it differs fundamentally from Expert Systems (ES) and Executive Information Systems (EIS) in its core purpose, target audience, and technology structure:

- **Decision Support Systems (DSS):** Designed for tactical and middle management to solve semi-structured problems using interactive data modeling and simulation tools, leaving final choices to the user.
- **Expert Systems (ES):** Built on artificial intelligence and a specialized knowledge base to mimic human domain experts. They use an inference engine to apply logical rules to specific scenarios, delivering concrete, automated recommendations or direct answers to problems.
- **Executive Information Systems (EIS):** Tailored specifically for senior executives to track long-term corporate performance. They emphasize tracking internal and external KPIs via high-level graphical dashboards, tracking exceptions, and drill-down tools, rather than complex predictive mathematical modeling.

(b) Write brief notes on Data Mining, Augmented Reality and Enterprise Resource Planning (ERP)

Data Mining: Data Mining is the systematic process of discovering hidden, mathematically significant patterns, correlations, and anomalies within massive corporate data repositories using advanced statistical algorithms, machine learning, and database management systems. It moves beyond standard data reporting to answer complex, predictive business questions, such as identifying shifts in customer buying habits or detecting fraudulent credit card transactions. By uncovering non-obvious relationships in data, data mining enables enterprises to forecast market trends, optimize pricing strategies, and build highly targeted marketing campaigns.

Augmented Reality (AR): Augmented Reality is an immersive digital technology that overlays computer-generated sensory content—such as 3D graphics, text, or animations—onto a user's real-world environment in real time. Unlike Virtual Reality (VR), which creates a completely simulated space, AR enhances the physical world with interactive digital information, viewed through smartphones, tablets, or smart glasses. In modern business, companies use AR for specialized employee training, remote maintenance tasks, interactive product visualizations, and virtual retail shopping experiences, blending digital precision directly into physical workflows.

Enterprise Resource Planning (ERP): Enterprise Resource Planning (ERP) refers to integrated, enterprise-wide software suites designed to consolidate and coordinate all core business functions—including finance, human resources, manufacturing, supply chain management, and sales—into a single centralized database system. By replacing isolated departmental systems with a unified digital ecosystem, an ERP ensures data flows seamlessly across the entire organization in real time. This single data source eliminates duplicate entries, standardizes operational workflows, and provides management with total cross-functional visibility to drive efficiency, agility, and scale.